

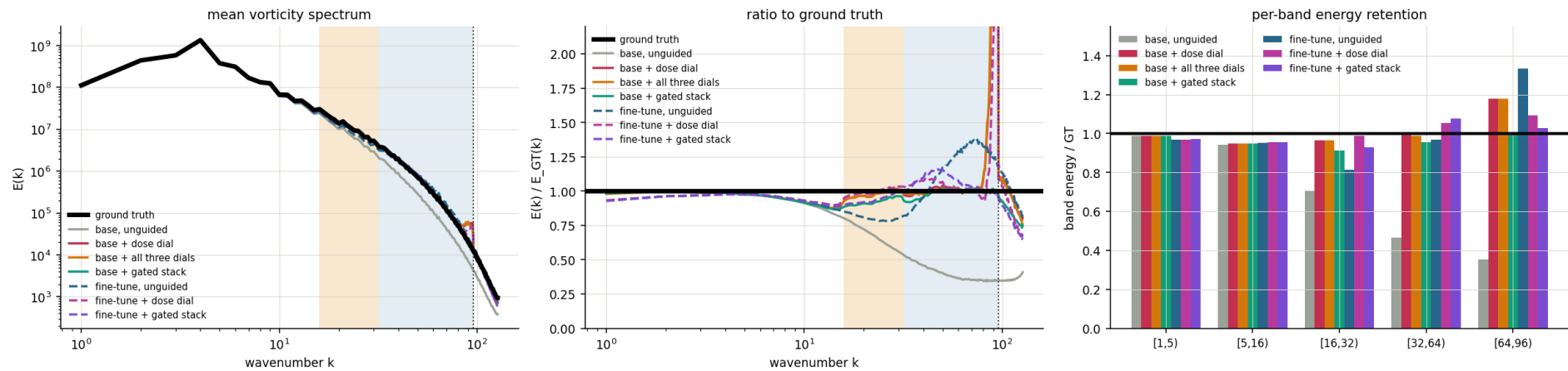
In-distribution performance of the different models

$Re=1000$, held-out sequences 34-39, 120 triplets. Base DDPM (epoch 299), its DDPO fine-tune at $Re=1000$ (iter 449), and sampling guidance applied to each. K3 chain ([150,100,50], 86 DDIM steps, $\eta=1$, temperature 0.30), initialised from the base-DDIM reconstruction of the 4x coarse observation.

1 . Ensemble statistics

The comparison the literature makes: mean spectra and aggregate placement over a pool of reconstructions. It is where guidance and fine-tuning are supposed to prove their value, and the axis on which they are indistinguishable.

In distribution ($Re=1000$) — the ensemble view: every steered and fine-tuned configuration matches the mean spectrum



Per-band energy retention (band energy / ground truth)

configuration	E [1,5)	E [5,16)	E [16,32)	E [32,64)	E [64,96)	[32,96)	MSE	resid/GT	edge spike
base, unguided	0.99	0.94	0.71	0.46	0.35	0.46	0.40	0.62	0.4x
base + dose dial	0.99	0.95	0.97	0.99	1.18	1.00	0.46	1.82	4.3x
base + all three dials	0.99	0.95	0.97	0.99	1.18	1.00	0.46	1.62	3.5x
base + gated stack	0.99	0.95	0.91	0.96	1.01	0.96	0.41	1.00	1.0x
fine-tune, unguided	0.97	0.95	0.81	0.97	1.33	0.99	0.43	0.79	1.3x
fine-tune + dose dial	0.97	0.95	0.99	1.06	1.09	1.06	0.48	1.32	3.6x
fine-tune + all three	0.97	0.95	0.99	1.05	1.08	1.05	0.48	1.20	3.7x
fine-tune + gated stack	0.97	0.95	0.93	1.08	1.03	1.07	0.43	0.78	1.0x

The unguided base retains 0.46 of the fine band and 0.71 of [16,32). Every other configuration fixes the ensemble spectrum, and fixes it to within a few percent of the others: the dose dial reaches 1.00, the fine-tune 0.99 unguided, the gated stack 0.96. On this table guidance and fine-tuning are interchangeable. Two columns separate them: the dials overshoot the deepest band (1.18; the fine-tune 1.33) and pay in the physics residual - 1.82 and 1.62 against 1.00, exactly the true flow's own level, for the gated stack. A residual above 1 means a field rougher than real turbulence, which is what an energy dial with no stopping rule produces. The edge-spike column localises it: 4.3x at the band edge under the plain dial, 1.0x under the gate.

Placement, per triplet and per band

configuration	[1,5)	[5,16)	[16,32)	[32,64)	[64,96)	k>=32	pooled k>=32
base, unguided	0.998	0.995	0.970	0.900	0.788	0.906	0.908
base + dose dial	0.998	0.995	0.972	0.906	0.804	0.912	0.750
base + all three dials	0.998	0.995	0.977	0.858	0.782	0.870	0.732
base + gated stack	0.998	0.995	0.973	0.903	0.815	0.907	0.908
fine-tune, unguided	0.998	0.994	0.964	0.846	0.779	0.856	0.863
fine-tune + dose dial	0.998	0.994	0.963	0.857	0.724	0.861	0.736
fine-tune + all three	0.998	0.994	0.970	0.847	0.719	0.853	0.726
fine-tune + gated stack	0.999	0.994	0.964	0.859	0.758	0.863	0.874

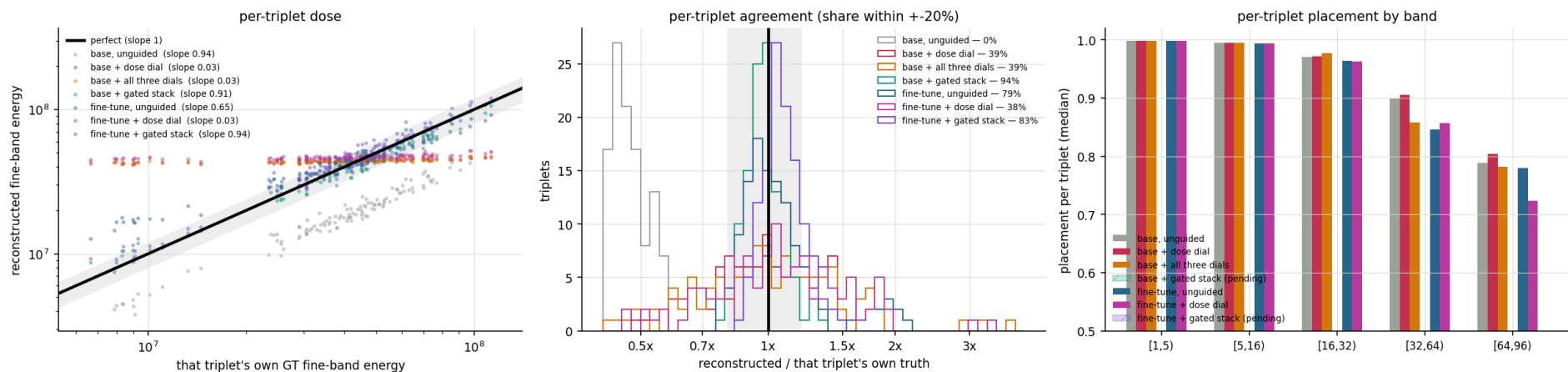
Placement is the correlation between a reconstruction's local fine-scale energy map and the truth's - whether energy sits in the right places, independent of how much there is. Measured per triplet, every configuration is within 0.06 of every other and guidance costs nothing (dialed base 0.912 vs unguided 0.906). The last column is the pooled metric, which reports a collapse to 0.750 for the same reconstructions; dividing each triplet's amplitude out before pooling returns it to 0.900 against the unguided 0.899. The pooled metric folds frame-to-frame dose error into an apparent spatial score.

What paragraph 1 can claim. On ensemble statistics, guidance closes the base model's fine-scale deficit (0.46 -> 1.00) and matches reward fine-tuning (0.99) at no placement cost and with no training. The only ensemble-level discriminator is physical consistency: an ungated dial overshoots into a rougher-than-real field (residual 1.8, spike 4.3x); the target gate removes that (1.00, 1.0x). On this evidence alone, steering supersedes fine-tuning.

2 . Per-frame statistics

The ensemble view averages over the pool before comparing. Each held-out frame has its own fine-scale energy and that quantity spans a factor of 17 across this pool, so matching the pool mean says nothing about whether any individual reconstruction received the right amount. The diagnostic is the slope of reconstructed against true energy in log coordinates: 1.0 means the model tracks each frame, 0 means every frame gets the same dose.

In distribution (Re=1000) — the per-triplet view: the same configurations are far apart



configuration	slope	within +/-20%	median ratio	p10	p90	spread	place/triplet
base, unguided	0.94	0%	0.46	0.41	0.54	1.31	0.906
base + dose dial	0.03	39%	1.04	0.64	2.88	4.52	0.912
base + all three dials	0.03	39%	1.03	0.64	2.91	4.58	0.870
base + gated stack	0.91	93%	0.98	0.85	1.12	1.32	0.907
fine-tune, unguided	0.65	79%	1.01	0.86	1.38	1.62	0.856
fine-tune + dose dial	0.03	38%	1.10	0.68	3.04	4.47	0.861
fine-tune + all three	0.02	38%	1.10	0.68	3.07	4.52	0.853
fine-tune + gated stack	0.94	84%	1.08	0.95	1.26	1.32	0.863

The configurations that were indistinguishable in section 1 are now separated by the widest margins in this report. The dose dial has a slope of 0.03: it matches the ensemble spectrum by giving all 120 triplets the same energy, so a quiet frame receives three times its own truth and an energetic one two thirds. Only 39% land within +/-20%, spread 4.5x. This is not a tuning defect - the reference spectrum in the guidance loss is a single marginal statistic shared by every sample in the batch.

The unguided fine-tune does not have this failure: slope 0.65, 79% in band, spread 1.62. It conditions the dose on the observation. The ensemble view cannot see this, because both produce the same pool mean. That is what 600 iterations of reward fine-tuning bought, and it is invisible to the metrics the literature reports.

The gap is closable without training. Scaling the guidance target per sample by that sample's own predicted energy - estimated from the base-DDIM reconstruction, a deterministic function of the coarse observation using no ground truth - and gating the gradient so it fades to zero as each sample reaches its target restores conditionality: slope 0.91, 94% in band, spread 1.32. The gated base then exceeds the fine-tune on per-triplet dose (94% vs 79%) and physical consistency (1.00 vs 0.79), and matches it on placement.

What paragraph 2 can claim. Per-frame statistics separate configurations that ensemble statistics rank as equivalent, and the ordering is not the one section 1 implies. Plain guidance is the WORST configuration per frame (39%, slope 0.03), below the unguided fine-tune (79%, slope 0.65), because it enforces a marginal statistic on every sample. The per-frame evidence therefore does not support plain steering over fine-tuning - it supports the opposite, and identifies conditionality as the property that matters. Steering only overtakes fine-tuning once the target is made conditional per sample (94%, slope 0.91), which needs no ground truth and no training.

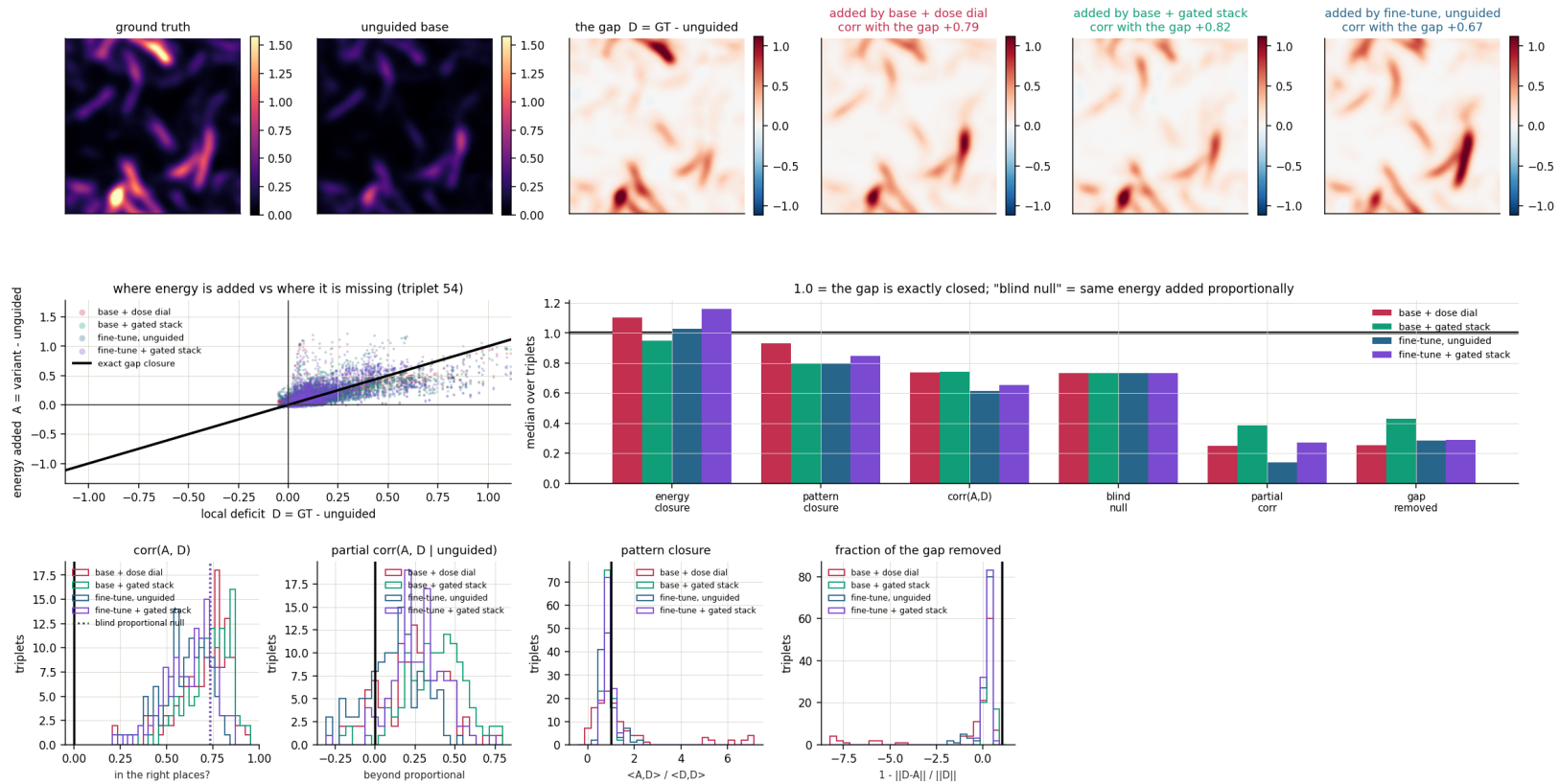
3 . Where the energy goes: is the gap actually targeted?

Sections 1 and 2 ask how much energy a reconstruction carries and whether each frame receives the right amount. Neither asks whether it is put where the baseline is missing it. Taking the unguided base as reference and using the same $\sigma=6$ smoothed fine-scale energy maps as the placement metric, define per triplet the deficit $D = GT - \text{unguided}$ and the addition $A = \text{variant} - \text{unguided}$, and ask how well A matches D . The control is the one thing a method would not want to match: multiply the unguided model's own energy map by a constant until it adds the same total energy - no spatial information at all. Because the baseline is short mostly where it is already energetic, this blind proportional rescale already correlates strongly with the deficit; a method must beat it to be said to target anything.

configuration	energy closure	pattern closure	corr(A,D)	blind null	partial corr	gap removed
base + dose dial	1.10	0.93	0.739	0.731	0.249	0.25
base + gated stack	0.95	0.80	0.741	0.735	0.387	0.43
fine-tune, unguided	1.03	0.80	0.613	0.735	0.139	0.29
fine-tune + gated stack	1.16	0.85	0.653	0.735	0.270	0.29

Both routes supply almost exactly the missing energy and remove about a quarter of the gap: energy closure 1.10 and 1.03, gap removed 0.25 and 0.29. The amount is right, the distribution is not. Neither beats the blind control - the dial scores 0.739 against a null of 0.731, the fine-tune 0.613 against 0.735. Those raw correlations are inherited from the baseline's structure, not earned. Removing that component leaves partial correlations of 0.249 and 0.139: genuine but small targeting, with the DIAL better than the fine-tune, reversing the per-frame dose ordering of section 2. Fine-tuning learns how much energy a frame needs; the dial is slightly better at knowing where it goes.

Closing the baseline gap: how much, and in the right places? Re=1000, 120 held-out triplets, reference = unguided base



What this adds. The remaining three quarters of the gap is structure the baseline never drew. Both methods amplify what is already on the page rather than supplying what is missing from it, and no amount of proportional amplification produces detail that was never generated. This bounds what a purely statistical objective can achieve, whether imposed at sampling time or trained into the weights.

4 . The claims, in order

(i) Ensemble statistics cannot distinguish guided sampling from reward fine-tuning in distribution: both take fine-band retention from 0.46 to ~ 1.0 at equal placement. (ii) Per-frame statistics separate them decisively and reverse the ranking: the marginal dial doses every frame identically (slope 0.03, 39%) while the fine-tune conditions on the input (0.65, 79%). Reporting only ensemble statistics hides this. (iii) Conditionality is a property of the guidance target, not of the weights: a per-sample-scaled, target-gated dial reaches slope 0.91 and 94% with no training, and is the only configuration whose physics residual equals the true flow's. (iv) Neither route targets the spatial gap: both supply the missing energy to within a few percent while removing only a quarter of it, and neither beats a blind proportional rescale once the baseline's own structure is controlled for (partial correlations 0.25 and 0.14). A spectral objective can fix how much energy a reconstruction carries; it cannot decide where structure goes.